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Evaluating EduTrack System Quality Using ISO/IEC 25010 and the DeLone and McLean Model

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ABSTRACT

The integration of information systems in higher education enhances efficiency, transparency, and academic service quality. However, continuous evaluation is crucial to ensure that such systems meet user expectations and comply with international quality standards. This study evaluates the quality of the EduTrack Information System developed for the Master's Program in Educational Research and Evaluation at Universitas Negeri Makassar. Using a quantitative descriptive method, the research integrates the ISO/IEC 25010 software quality standard and the DeLone & McLean Information System Success Model. Data were collected from 60 active student users through questionnaires and technical testing with tools such as Apache JMeter, OWASP ZAP, Mozilla Observatory, K6, and the System Usability Scale (SUS). EduTrack demonstrated excellent results in functional suitability (100%), reliability (96.55%), and usability (SUS score = 71.5, good). The PLS-SEM analysis confirmed that service quality (β = 0.580) and information quality (β = 0.390) significantly affect user satisfaction (R^2 = 0.716). These findings indicate that EduTrack performs effectively and aligns with user expectations, though security enhancements are still required. This study demonstrates an integrated evaluation framework that combines technical quality standards and user satisfaction analysis, providing a reference model for future assessments of academic information systems.

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INTRODUCTION

Information systems play a pivotal role in managing study programmes in higher education institutions. Integrated information systems can facilitate the efficient and transparent management of academic, administrative, and human resource data. This aligns with previous studies indicating that information systems enhance data accuracy and efficiency while facilitating better decision-making within educational organisations (Frisdayanti 2019). One crucial aspect of information systems in program management is their capacity to support the planning and evaluation processes. The effective implementation of such systems not only increases operational efficiency but also strengthens collaboration among stakeholders in educational institutions. With an integrated platform, information becomes easily accessible to lecturers, students, and program administrators, fostering a more collaborative and responsive academic environment [1].

As part of efforts To improve transparency, efficiency, and quality of academic services, EduTrackwas developed as an academic information system for the Master's Program in Educational Research and Evaluation (S2 PEP) at Universitas Negeri Makassar. This application provides several key features, including program information, academic news, satisfaction questionnaires for lecturers and students, tracer studies, and thesis completion monitoring. Through EduTrack, students and lecturers can easily access the latest academic information,

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provide feedback on service quality, and track their study progress. However, despite its numerous benefits, continuous evaluation of the quality and effectiveness of the system remains essential. Such evaluations ensure that the system meets user expectations and complies with established quality standards. Identifying both the strengths and weaknesses of the system is vital for providing improvement recommendations that enhance system performance and user satisfaction [2] [3].

In the higher education context, evaluating information systems is a critical step to ensure that they not only improve administrative efficiency but also contribute to enhancing the overall quality of academic services. Periodic evaluation allows institutions to align their systems with evolving needs and maintain relevance in supporting academic goals [4] [5]. One widely used framework for evaluating information systems is the DeLone and McLean Information Systems Success Model, which emphasises system quality, information quality, and *user* satisfaction as key indicators of success [6]. This model has been extensively applied across various sectors, including education, owing to its ability to provide comprehensive insights into system effectiveness and user experience.

The DeLone and McLean model identifies six interrelated variables: system quality, information quality, service quality, use, user satisfaction, and net benefits [7]. By employing this model, institutions can assess how well an information system meets user needs and delivers the expected benefits. Among these variables, user satisfaction serves as a key indicator of how satisfied users are with the system. Empirical studies have shown that system and service quality significantly influence user satisfaction, particularly in academic service systems. A study conducted at the Faculty of Economics, Universitas Negeri Jakarta, revealed that user satisfaction with academic service systems was strongly affected by both system and service quality [8]. Consequently, this study focuses on these two variables, system quality and service quality, to measure user satisfaction.

In addition to the DeLone and McLean model, international standards such as the ISO 25010 play an essential role in objectively and systematically assessing software quality. This standard provides a comprehensive framework for evaluating software products by identifying relevant characteristics and subcharacteristics [9]. ISO 25010 supersedes the earlier ISO 9126 standard and offers a more integrated approach to assessing software quality. The ISO 25010 standard categorises software quality into eight interrelated characteristics: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. By integrating the DeLone and McLean model with the ISO 25010 standard, this study aims to comprehensively evaluate the quality of the EduTrackinformation system. The evaluation will not only identify the system's strengths and weaknesses but also provide strategic recommendations for enhancing the quality of technology-based academic services. Thus, the findings of this study are expected to contribute to the development of academic information systems that are more effective, efficient, and aligned with users' needs.

METHOD

This study uses a descriptive quantitative approach with an information system evaluation method based on the ISO/IEC 25010 standard and the DeLone and McLean Information System Success Model. This study combines two perspectives: the Technical Perspective (ISO 25010), which assesses software quality based on eight main characteristics, and the user perspective (D&M IS Model), which assesses the relationship between system quality, quality of service, and user satisfaction. The research was conducted using the EduTrackInformation System (https://s2pep-unm.com) used by Makassar State University students. The research procedure is shown in Figure 1.

Several previous studies have integrated the ISO/IEC 25010 and DeLone & McLean models within the context of information system evaluation, particularly for e-learning and academic

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portals [10][11]. However, most implementations of these models have still been conducted separately between technical software testing and user-perception assessment. This study differs in that it explicitly combines both approaches into a single integrated analytical framework to evaluate a postgraduate-level academic information system, namely EduTrack. Through this approach, the study not only assesses the technical quality of the software based on the international ISO/IEC 25010 standard, but also examines the causal relationships among system quality, service quality, and user satisfaction using Partial Least Squares–Structural Equation Modeling (PLS-SEM).

The hybrid model developed in this research provides a more comprehensive perspective on information system success by simultaneously incorporating both objective and subjective dimensions. Conceptually, the study extends the combined application of ISO/IEC 25010 and DeLone & McLean—previously limited to e-learning contexts—into the domain of postgraduate academic information systems, which possess different complexities and characteristics. Accordingly, this research offers novelty through empirical validation of an integrative two-model approach, while also providing practical contributions in the development of a more holistic evaluation framework that can be replicated by other higher-education institutions.

Scientifically, the contributions of this study lie in three main areas. First, it proposes an integrated evaluation model combining ISO/IEC 25010 and DeLone & McLean, capable of bridging the gap between technical quality assessment and user experience. Second, the study validates this model using real empirical data from EduTrack system users in a higher-education environment through multivariate statistical analysis. Third, the research presents an applicable and measurable approach for educational institutions seeking to enhance the governance and quality of academic information systems, grounded in international standards and end-user perceptions.

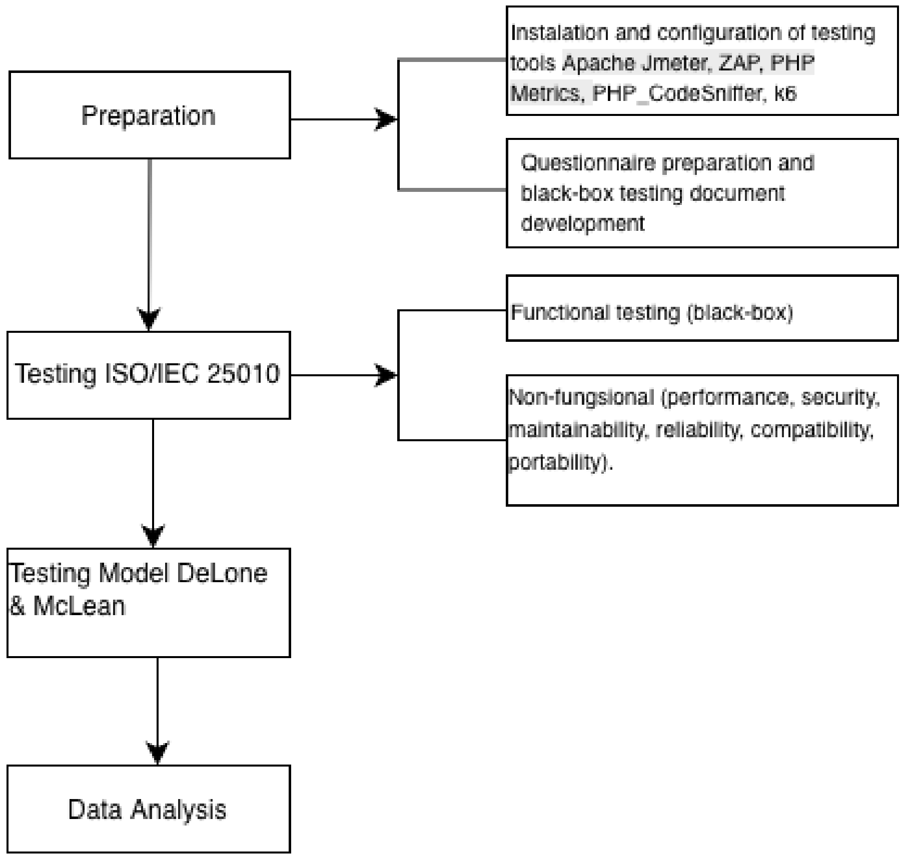

Figure 1. Research Prosedure

A. Research Variables

1. Variables Based on ISO/IEC 25010

The quality assessment of information systems was conducted with reference to the ISO/IEC 25010 standard, which is an international framework for evaluating software quality. This standard contains eight main characteristics of system quality, each of which consists of several sub-characteristics that are measurement indicators. However, in this study, not all variables and sub-characteristics of the indicators were used. Some Variables in ISO 25010 testing were not used for the following reasons:

1) Portability testing was not performed because this information system runs in the browser; therefore, there is no need for adaptation testers on various local devices and OS. This is supported by research showing that portability tends to be more relevant for installed software than for intrinsically portable web-based applications [12], [13].
2) Maintainability testing often depends on the context of the development and operation of the system, which in many cases is not very relevant for education systems that have shorter life cycles and are frequently updated [14], [15].

Furthermore, the selection of sub-characteristics was carried out purposively by considering the relevance to the context of the EduTrackinformation system and the availability of representative testing tools. Other sub-characteristics that are not used are considered insignificant to the purpose of the study because they have no direct effect on the user's performance and satisfaction. Here are the six quality variables tested along with specific test sub-indicators and focuses, as well as methods or tools used for testing.

1) Functional Suitability includes the sub-indicators of Completeness and Correctness. This test was carried out using the Black Box Testing approach with the test case method by one expert judge. Each test case was evaluated using the Guttman scale, with successful or unsuccessful responses. Test case testers through expert assessment of their efficiency and effectiveness. According to Prakash, well-constructed test cases help uncover critical errors within the software system and are instrumental in managing the complexities associated with software testing [16].
2) Performance Efficiency includes the sub-indicators of time behaviour, resource utilisation, and capacity. The focus of this test was to measure the speed and efficiency of the system. The tools used in this test included Apache JMeter version 5.6.3.
3) Security was tested using the sub-indicators of Confidentiality, Integrity, and Authenticity. Its main focus is to maintain data security and control access to the system. The tests were conducted using the Zed Attack Proxy (ZAP) and Mozilla Observatory.
4) Reliability includes aspects of maturity and availability, the purpose of which is to test the resilience of the system under various conditions. The tool used for this test is K6.
5) Compatibility tests system compatibility through the Cross-Browser and Cross-Device sub-indicators. This test aims to ensure that the system can run smoothly across different browsers and devices. The tools used included PowerMapper SortSite, Google Chrome DevTools, and Browserling.
6) Usability focuses on the sub-indicators of Learnability, Operability, and Appropriateness. The purpose of this test was to assess the user's ease of use of the system. The method used was a System Usability Scale (SUS) questionnaire with 25 student respondents who had used the

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EduTrackinformation system. The SUS is the most widely used standardised questionnaire for assessing perceived usability [17].

2. Variables Based on DeLone & McLean

The variables used based on the DeLone & McLean Information System Success model are system quality, service quality, and user satisfaction. The purpose of this test was to assess the quality of the EduTrackinformation system from the perspective of the user based on System Quality, and Service Quality. The selection of these two variables represents the technical dimension (System Quality) and the support dimension (Service Quality) which collectively determine how satisfied users are when interacting with information systems [8]. The relationships between the variables are shown in Figure 2.

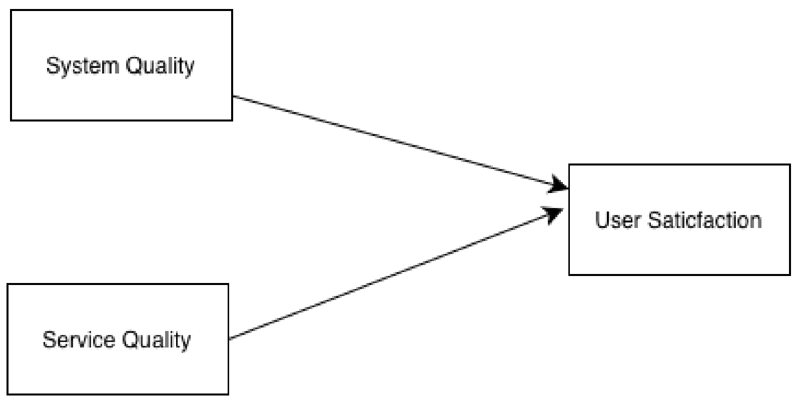

Figure 2. Relationship between DeLone & McLean variables

The data collection process was carried out through the distribution of questionnaires to 60 students who actively used the EduTrackinformation system. The questionnaire was designed using a five-point Likert scale (1–5), in which respondents were asked to rate several indicators representing the main variables in the DeLone and McLean models, namely:1) Service Quality – measures the quality of support and service provided by the system administrator to users [6]; 2) System Quality – assesses the reliability, ease of use, and technical performance of the EduTracksystem; and 3) User Satisfaction – measures the level of student satisfaction after using the system in their academic activities.

The data of this study was obtained from 60 active students who have used the EduTrack information system. The determination of the sample count was based on the minimum limit recommended in the Partial Least Squares–Structural Equation Modeling (PLS-SEM) analysis. According to [18] the PLS-SEM method is suitable for use in exploratory research and is able to produce stable estimates even with a relatively small sample number, which is between 30 and 100 respondents depending on the complexity of the model. In this study, the structural model involved three latent variables and several indicators, thus meeting the rule criteria 10 times, i.e. the minimum number of samples is equal to ten times the maximum number of structural paths leading to a construct. Therefore, the use of 60 respondents is considered adequate to achieve sufficient statistical strength and represent the active user population of the EduTrack system at Makassar State University. The selection of respondents was carried out purposively, with the criteria of students who actively use the system in academic activities such as filling out questionnaires, registering courses, and monitoring thesis completion. This approach ensures that all respondents have relevant experience and can provide a valid assessment of the quality of the system and services.

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B. Data Analysis Techniques

1. ISO/IEC 25010

a. Functional Suitability

The data from the expert judgment assessment were calculated using the following formula:

$$X = \frac{I}{P} \text{ x } 100\%$$

Where:
I=Number of successful functions (value 1)
P= Total number of functions designed

Furthermore, the results can be interpreted according to the criteria in Table 1 as follows.

Table 1. Interpretation Description Functional Suitability

No	Persentage of Feasibility	Interpretation Description
1	0%-20%	Very Bad
2	21%-40%	Poor
3	41%-60%	Fair
4	61%-80%	Good
5	81%-100%	Very Good

Source:[19]

b. Usability

Usability measurements were performed using the SUS instrument, which is a standard software questionnaire containing 10 items filled out by the user after interacting with the system being tested. Each item used a Likert scale with a score of 1-5. A score of 1= strongly disagrees and 5= strongly agrees. Once the responses are collected, the score of each item is changed through a specific counting procedure: for odd-numbered (positive) questions minus 1, while for even-numbered (negative) questions, the score is given as 5 minus the response value. The sum of the scores was then multiplied by 2.5 to give a total score ranging from 0 to 100 [17]. The interpretation of the SUS category is presented in Table 2.

Table 2. SUS Category Interpretation

Category	Range	Score
Acceptability	Not Acceptable	0-50
Ranges	Marginal-Low	51-62
	Marginal-High	63-70
	Acceptable	71-100
Adjective Ratings	Best Imaginable	86-100
	Excellent	81-85
	Good	71-80
	OK	50,9-70
	Poor	26-50,8
	Worst Imaginable	0-25

Source [21]

2. DeLone and McLean

At this stage, the *partial least squares structural equation modelling* (PLS-SEM) model was tested to evaluate the relationships between the constructs used in the DeLone and McLean Information Systems Success Model. This analysis aims to understand how *System Quality*, *Service Quality*, and *User Satisfaction affect* each other in the context of using the EduTrackinformation system. The relationships between the variables are shown in Figure 2.

a. Outer Model Test

An *outer model* test was conducted to assess the validity and reliability of the indicators representing each construct. Some of the steps taken include the following:

a) The convergent validity test was conducted by examining the loading *factor value* of each indicator against its construct. An indicator is declared valid if it has a *loading factor* value of more than 0.5, indicating that the indicator can adequately explain the construct.
b) The construct reliability test was conducted using two main measures: a) Composite Reliability (CR) which describes the internal consistency of the indicator, with the criterion of CR value > 0.7; and b) Average Variance Extracted (AVE), which indicates the extent to which the construct is able to explain the variance of the indicators, with a minimum AVE value of > 0.5.

b. Inner Model Test

After the measurement model was declared valid and reliable, *an internal model test* was conducted to evaluate the relationship between the constructs in the structural model.

a) The analysis was carried out by examining the path coefficient value between latent variables to determine the direction and strength of the relationship between constructs.
b) The value of R^2 (R-square) is used to determine the magnitude of the proportion of the variance of the dependent construct described by the independent construct. The higher the R^2 value, the better the model's ability to explain the relationships between the variables.

RESULTS AND DISCUSSION

A. ISO/IEC 25010 Testing

1. Functional Suitability

Functional Suitability testing is performed by a single expert using the black-box testing approach, which focuses on checking the functioning of the system without looking at the source code. This test aims to ensure that every feature of the UNM EduTrackInformation System has run according to the needs of users and the specifications that have been set. The testing process began with the identification of the main functions of the system, such as the login process, student data management, document uploading, academic report creation, and printing evaluation results. Each of these functions was then tested based on *the expected inputs* and *outputs* using a pre-prepared test scenario in the *test case* document. The number of Use Cases in this test was 20, with 75 test cases. The test results showed that the number of successful scores was 75, with the number of functions tested being 75. Therefore, the total score obtained was (75/75) × 100% = 100%. The test results show a 100% success rate in all test scenarios; therefore, the EduTracki information system can be categorised very well in terms of Functional Suitability.

2. Performance Efficiency

The focus of the test on performance efficiency was to measure the speed and efficiency of the EduTrackinformation system. In the time behaviour aspect, the test focuses on the speed at which the system responds to user requests. Testing was conducted on two main features using Apache Jmeter software, with each feature involving ten simultaneous users. The results of the login process test had a response time of 281 ms, while the report generation process showed a response time of 267 ms with an error rate of 0%. The test results showed a response time of less than 300 ms, which was in the very good category. This is in line with Dako, who stated that an ideal response time of under 2 seconds is considered excellent, while a time between 2 and 5 seconds can still be categorised as adequate, although it can lead to dissatisfaction if it exceeds that limit [19].

Furthermore, for resource utilisation, testing is performed to measure the efficiency of the use of hardware resources, such as the CPU and RAM, when the system is operating under normal load conditions. The results show that the CPU usage is below 1% and that approximately 1.6 GB of RAM is used when four browsers are active simultaneously. These findings indicate that the system works efficiently and does not overload the device, thus exhibiting a good level of resource efficiency. On the sub-characteristic of capacity. The test was aimed at measuring the system's ability to handle a simultaneous number of users. The results showed that the system could process 80 logged-in users within 10 s without any significant failures or performance degradation. This indicates that the system is stable and scalable and can maintain its performance despite an increase in the number of users. Overall, the test results show that the system has excellent Performance Efficiency. Fast response times, efficient use of resources, and high user handling capacity are indicators that the system has met the ISO/IEC 25010 quality standards in terms of performance.

3. Security

Security testing was performed using two main tools: OWASP ZAP and Mozilla Observatory. Based on the results of the analysis with OWASP ZAP, 19 alerts were found, consisting of one high, four medium, and eight low categories. The inspection results using the Mozilla Observatory showed a security score of 5/100 (Grade F), indicating that the security of the system is relatively low and needs to be improved. In the Confidentiality sub-indicator, the detected system displayed the server version header openly, and the cookies were not protected with security attributes such as Secure, HttpOnly, and SameSite. This condition is classified as vulnerable because it can be used by unauthorised parties to carry out session hijacking or information disclosure attacks. The recommendation is that developers hide the server version from the HTTP header and enable Secure Flags on cookies to ensure that data is only sent over encrypted connections. Furthermore, in the Integrity subindicator, the test results show the absence of anti-CSRF tokens and the lack of implementation of important HTTP security headers such as content security policy (CSP), HTTP Strict-Transport-Security (HSTS), and X-frame-options (XFO). This condition is categorised as weak because it opens up the potential for cross-site request forgery, clickjacking, and data injection attacks. The main recommendation is to add a CSRF token to each authentication form and implement OWASP standard security headers to maintain data integrity during information exchange.

Meanwhile, in the Non-Repudiation sub-indicator, it was found that the trail audit mechanism was not yet available in the system. This means that user activity (such as logins, uploads, or data changes) has not been systematically recorded. This condition is limited because the system cannot provide digital evidence to track who performed certain actions. The recommendation is for developers to add a feature for logging user activity (transaction logging) that includes user identity, time, and activity type. Overall, the test results show that the security aspects of the EduTracksystem still need to be improved, especially in session management, security headers, and activity logging. The implementation of these recommendations will improve the security score and strengthen the fulfilment of security characteristics within the framework of ISO/IEC 25010.

4. Reliability

Reliability testing was performed using the K6 Load Testing Tool and GitHub action workflow. The maturity sub-indicator test was carried out using *the K6 Load Testing Tool* with 40 virtual users (VU) for 1 min. The results showed a total of 2,178 tests with a success rate of 96.55%, failure rate of 3.44%, and average response time of 616 ms. No crashes or significant performance degradation were observed during the tests. This value indicates that the EduTrackinformation system has an excellent level of stability and can maintain service consistency under the simultaneous load of users. Thus, the *Maturity* indicator is declared to be very good (96.55 %), in accordance with the Reliability criteria in ISO/IEC 25010. Furthermore, the availability sub-indicator was tested for 24 h with execution every 1 h with optimistic testing using GitHub Actions Workflows. Each workflow executed an HTTP request script to the system's primary address (https://s2pep-unm.com) to retrieve the response code state and server response time. During the test period, all 24 workflows were successfully executed without errors or access interruptions. Each run received an OK 200 HTTP code response with an average response time of between 7 and 12 s and no connection failures (timeouts or 5xx errors). These results show that the system operated stably during the test period, with an availability rate of 100% (24 h nonstop).

5. Compatibility

The results of the system compatibility test show that the EduTrackinformation system has an excellent level of compatibility with various browsers and devices. Based on Cross-Browser testing, the appearance of the EduTrackinformation system was consistently displayed on popular browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Brave, without any significant differences in appearance or functionality. This indicates that the system was designed with good web standards so that it could operate stably on various platforms. Furthermore, the Cross-Device test showed that the website display ran optimally on laptop and iPad devices, but there was a slight shift in the smartphone display, especially in the *navbar*. This issue is minor and can be fixed by customising the CSS code to make the layout fully responsive.

In addition, the test results using PowerMapper SortSite reinforce the conclusion that none of the pages experience compatibility issues with modern browsers, indicating that the HTML and CSS structure of the site is compliant with W3C standards. Testing with Google Chrome DevTools also showed that the display of the EduTrackinformation system was responsive at most screen resolutions, leaving only a slight visual improvement at certain resolutions. Meanwhile, tests with Browserling showed that the website can run well on all popular browser combinations on the Windows 10 operating system. Overall, the system is highly compatible with a wide range of browsers and devices. The website is stable and performs consistently in various environments, with only a slight refinement of the display aspect on smartphone devices to achieve full perfection in terms of *responsiveness* and user experience.

6. Usability

The respondents in this test were 25 students who had used the EduTrackinformation system. Based on the data processing results of 25 respondents, the individual SUS score was in the range of 55–92.5, with details as shown in Figure 3.

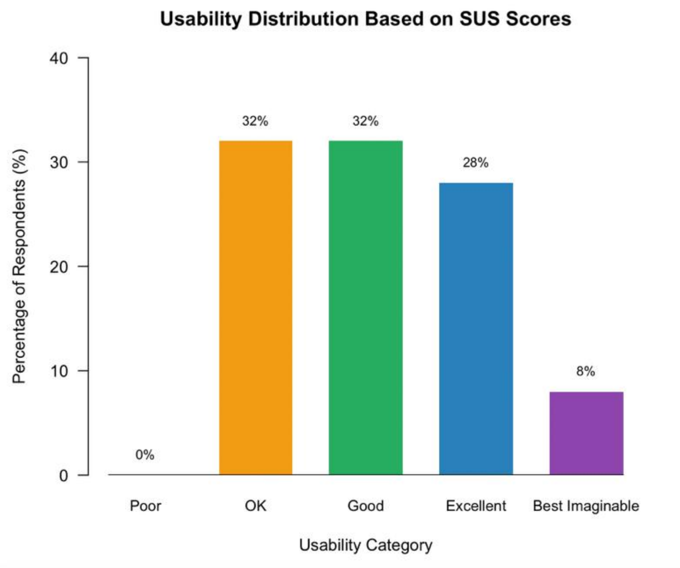

Figure 3. SUS test results

The average SUS score was 71.5, with the categories of Acceptability Ranges = Acceptable and Adjective Ratings = Good. The test results show that the users rated the system as having a good level of usability. The majority of respondents gave above-average scores, indicating that the system was easy to learn, intuitive, and did not cause excessive cognitive burden during operation.

However, some respondents scored below 60, indicating a need for improvement in certain aspects, such as interface consistency, navigation between pages, and icon clarity. This can be used as the basis for the next iterative development to make the system easier to operate by new and nontechnical users. Thus, based on the results of the calculation of SUS scores greater than 70, the EduTrackinformation system can be categorised as feasible in terms of usability and meets the minimum user acceptance standards for applications in health and education environments [20] [21].

B. Test Results Based on DeLone & McLean Model

Testing based on the success model of Delone and McLean information systems was conducted to see more thoroughly how user satisfaction perception of Edutrack's information system is reviewed from system and service quality. The data analysis method for this test used PLS-SEM with 60 respondents.

1. Outer Model Results

An Outer Model evaluation was conducted to test the validity and reliability of the constructs used in this study. This measurement model aims to ensure that the indicators representing each latent variable have good internal consistency, adequate convergent validity, and maintained discriminant validity. The results of the outer model test are shown in Figure 3, and a recap of the convergent validity test is presented in Table 3. Based on the test results, all indicators in the System Quality, Service Quality, and User Satisfaction variables showed a loading factor value above 0.70, so all indicators were declared valid and could be used in the next stage of analysis. The results of this test also show that each indicator strongly contributes to explaining the latent variables it represents.

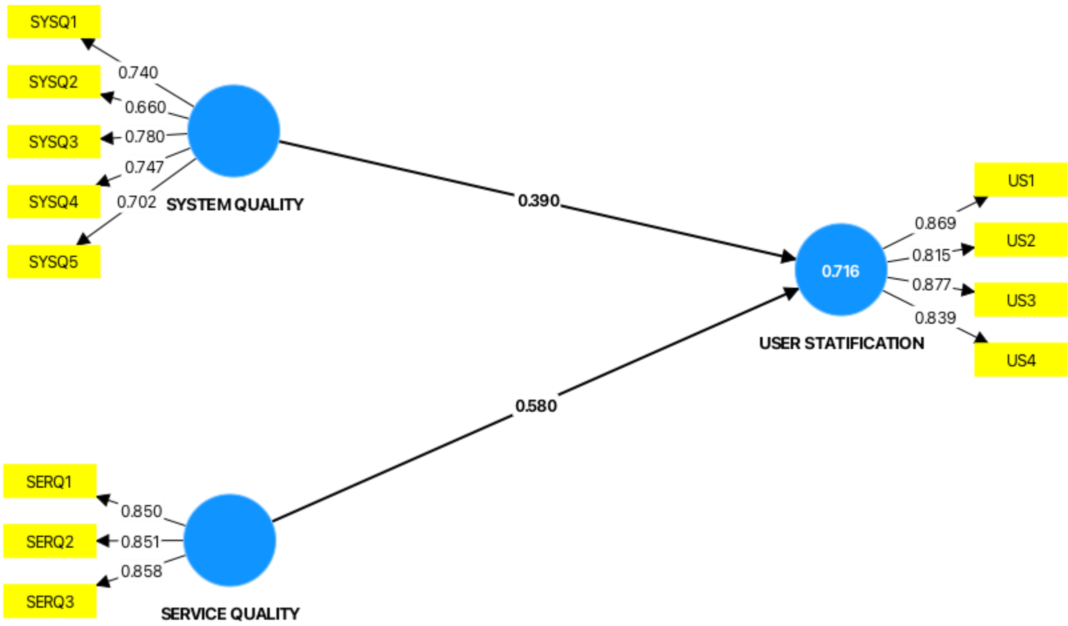

Figure 4: Result oof the Outer Model Assessment

Furthermore, the results of the reliability test can be seen in Table 4 of the Composite Reliability (CR) Test and Cronbach's alpha shows a value above 0.70, so it can be concluded that the construct has good internal reliability. This shows that the indicators of each variable have a high level of consistency in measuring the same concept. The results of the Average Variance Extracted (AVE) test showed values above 0.50 for each variable, indicating that the convergent validity criteria were met. Thus, it can be concluded that most of the variance in the indicator is explained by the latent construct in question.

Overall, the results of the Outer Model evaluation prove that all indicators meet the criteria of validity and reliability. Thus, the measurement model in this study is declared feasible to proceed to the Inner Model testing stage to see the relationship between latent variables in the structural model of the research.

Table 3. Outer Model Convergent Validation Results

Construct	Indicator	Loading Factor
System Quality	A1	0.609
	A2	0.613
	A3	0.791
	A4	0.553
	A5	0.649
Service Quality	B1	0.842
	B2	0.837
	B3	0.836
User Satisfaction	P1	0.816
	P2	0.788
	P3	0.858
	P4	0.824

Table 4. Outer Model Reliability Test Results

	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
Service Quality	0.813	0.815	0.889	0.727
System Quality	0.779	0.790	0.848	0.529
User Statification	0.872	0.874	0.913	0.723

2. Inner Model Results

The results of the Service Quality Relationship Test on User Satisfaction showed a path coefficient value of 0.580, t-statistic of 6.304, and p-value of < 0.000. The value exceeds the significance threshold of 1.96 (at α = 0.05); therefore, it can be concluded that Quality of Service has a positive and significant effect on User Satisfaction. In practical terms, improvements in service quality, such as response speed, reliability of technical support, and system providers' ability to handle user complaints, contribute significantly to improving user satisfaction levels. The coefficient value of 0.580 also indicates a relatively strong and substantial influence.

Furthermore, the results of the Information Quality Relationship Test on User Satisfaction produced a path coefficient of 0.390, a t-statistic of 3.495, and a p-value < 0.000. A t-value greater than 2.57 (α = 0.01) confirmed that this influence was statistically significant. This means that the quality of information generated by the system, such as accuracy, relevance, completeness, and ease of understanding, contributes to improving user satisfaction. An R^2 value of 0.716 indicates that 71.6% of the variation in User Satisfaction can be explained by the variables Service Quality and Information Quality, while the remaining 28.4% is explained by other factors outside the model. Based on Chin's (1998) guidelines, R^2 values above 0.67 are categorised as substantial, which indicates that the model has strong predictive capabilities against endogenous variables.

Overall, the PLS-SEM model in this study shows a positive and significant relationship between Service Quality and Information Quality on User Satisfaction. Both variables play an important role in explaining the level of user satisfaction with the system, with the largest contribution coming from QoS (β = 0.580). This confirms that responsive and professional service support is a key factor in the successful implementation of information systems.

The findings of this study not only validate the technical and service quality of EduTrack but also demonstrate its potential as a benchmark model for developing academic information systems in other universities. The integration of the ISO/IEC 25010 standard and the DeLone & McLean model provides a dual evaluation framework that balances objective system performance with user-centered perceptions. This combination allows institutions to design quality assurance mechanisms that are both measurable and user-responsive. Practically, EduTrack can serve as a reference for other higher education institutions aiming to modernize their information systems while maintaining compliance with international standards. By adopting the same hybrid evaluation approach, universities can identify weaknesses early, improve user satisfaction, and enhance institutional governance. Furthermore, the use of open-source tools such as Apache JMeter, K6, and OWASP ZAP makes this framework cost-effective and replicable, supporting continuous improvement across various academic environments.

This study has several limitations that should be acknowledged. First, the sample size consisted of 60 active users, which although sufficient for PLS-SEM analysis may limit the generalizability of the findings to larger populations. Future studies could expand the number of respondents or include multiple universities to strengthen comparative analysis. Second, the evaluation primarily focused on technical and user-perception indicators, without incorporating organizational and managerial aspects such as governance maturity or system adoption behavior. Subsequent research could integrate complementary models such as COBIT 5 or UTAUT to provide a broader understanding of system performance and acceptance. Finally, longitudinal studies are recommended to examine how system updates, policy changes, or user training affect long-term satisfaction and system sustainability.

CONCLUSIONS

The evaluation results confirm that the EduTrack system meets the ISO/IEC 25010 technical standards and demonstrates strong performance, reliability, and usability. Meanwhile, the DeLone and McLean model findings emphasize that service quality and information quality significantly influence user satisfaction, validating the system's overall effectiveness. To further enhance its contribution, future research could extend this hybrid evaluation framework to longitudinal studies examining user behavior and satisfaction trends over time or apply it to multiple higher education institutions for comparative analysis. Expanding the model to include governance and adoption dimensions such as those in COBIT 5 or UTAUT would also provide a more holistic understanding of academic information system success. Overall, the hybrid ISO/IEC 25010–DeLone & McLean evaluation framework proposed in this study can guide institutional decision-making and strategic planning in digital transformation initiatives across universities.

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